

M3 Project Report

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Abstract

Performance of All classification methods described in the class needs to be compared using the MNIST dataset. Classification methods covered in this report are: Logistic Regression, K Nearest Neighbors, Support Vector Machines, Decision Trees, Random Forest and Naive Bayes.

Classification Methods

Empirical results

Empirical results are as follows:

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RESULTS SUMMARY - WITHOUT PCA
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Classifier	Accuracy	Precision	Recall	F1-Score	Train Time (s)	Predict Time (s)
Logistic Regression	0.9350	0.9350	0.9350	0.9350	9.505056	0.005948
K Nearest Neighbors	0.9450	0.9450	0.9450	0.9450	0.005881	3.091135
Support Vector Machines	0.9675	0.9675	0.9675	0.9675	0.271566	0.133779
Decision Trees	0.7500	0.7500	0.7500	0.7500	0.186109	0.002781
Random Forest	0.9425	0.9425	0.9425	0.9425	0.227169	0.034196
Naive Bayes	0.7200	0.7200	0.7200	0.7200	0.043561	0.092582

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 **Key Insights:**

-  **Best Accuracy:** Support Vector Machines (0.9675)
-  **Fastest Training:** K Nearest Neighbors (0.0059s)
-  **Fastest Prediction:** Decision Trees (0.0028s)

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Fig 1. Classification Methods Performance Summary.

Logistic Regression

Pros:

- Strong Accuracy 93.5% competitive with KNN and Random Forest.
- Very Fast prediction
- Interpretable: Provides coefficients and probabilities, making decisions explainable.
- Good baseline

Cons:

- **Slowest training time.**
- **Moderate performance** on MNIST (93.5%)
- **Lower accuracy** than SVM.
- Assumes linear decision boundaries
- May underfit complex patterns
- Requires feature scaling

K Nearest Neighbors

Pros:

- **Fastest training time** 0.005 seconds.
- **Simple** and intuitive
- **High accuracy** on MNIST (94.5%)
- Adapts to local patterns: Uses local neighbourhoods, effective for varied class distributions.

Cons:

- **Very slow prediction time.**
- High memory requirements (stores entire dataset)

Support Vector Machines

Pros:

- **Highest accuracy** among classical ML methods (96.5%)
- **Fast Training** – much faster than Logistic Regression
- **Fast prediction time.**
- Handles non-linear patterns via kernel functions.
- Good generalization: Strong performance suggests same.
- Works well with clear margin of separation

Cons:

- **Slower training** than of the alternatives i.e. KNN, Decision Trees, Random Forest and Naïve Bayes.
- **Slower predictions** than fastest models Decision Trees, Logistic Regression and Random Forest.
- Harder to explain.
- Limited scalability: Training time can grow significantly with large datasets.

Decision Trees

Pros:

- **Fastest prediction time:** 0.0028 seconds — fastest among all models, suitable for real-time inference.
- **Fast training:** 0.1861 seconds — much faster than Logistic Regression (9.51s), competitive with other models.
- **Interpretable:** Visualizable tree structure; easy to explain decisions.
- **Feature importance:** Provides insights into which features matter most.

Cons:

- **Second Lowest accuracy:** 75.00% — worst among all models except Naïve Bayes.
- **Overfitting risk:** Despite max_depth=20, still shows lower generalization than ensemble methods.
- **High variance:** Sensitive to small data changes; Random Forest (ensemble) performs better (94.25%).
- **Struggles with high-dimensional data:** 784 features may lead to suboptimal splits.
- **Less effective than ensemble:** Random Forest (100 trees) achieves 94.25% vs 75.00% for a single tree.

Random Forest

Pros:

- **High accuracy:** 94.25% — second best after SVM (96.75%), better than Logistic Regression (93.50%) and Decision Trees (75.00%).
- **Fast training:** 0.23 seconds — much faster than Logistic Regression (9.51s), competitive with SVM (0.27s) and Decision Trees (0.19s).
- **Fast prediction:** 0.034 seconds — good for real-time use, faster than KNN (3.09s) and Naive Bayes (0.09s).
- **Handles high dimensionality:** Works with 784 features without dimensionality reduction.
- **Good balance:** Strong accuracy with reasonable training and prediction times.

Cons:

- **Lower accuracy than SVM:** 94.25% vs 96.75% — SVM performs better on this dataset.
- **Memory intensive:** Stores 100 decision trees, requiring more memory than single models.
- **Less interpretable than single models:** Harder to explain than Logistic Regression or a single Decision Tree.
- **Slower than some models:** Training (0.23s) is slower than KNN (0.006s) and Naive Bayes (0.04s).

- Not the fastest predictor: Prediction (0.034s) is slower than Logistic Regression (0.006s) and Decision Trees (0.003s).

Naive Bayes

Pros:

- **Very fast training:** 0.0436 seconds — among the fastest.
- Simple and interpretable: Probabilistic model with clear assumptions; easy to understand and explain.
- Works well with small datasets: Can perform reasonably even with limited training data.
- Low memory usage: Only stores mean and variance per feature per class.
- No hyperparameter tuning needed: Works out-of-the-box with default parameters.
- Probabilistic outputs: Provides class probabilities, not just predictions.

Cons:

- **Lowest accuracy:** 72.00% — worst among all classifiers.
- **Independence assumption:** Assumes features are independent, which is violated for image pixels (spatial correlations exist).
- **Slower prediction** than some.
- Poor fit for image data: Gaussian Naive Bayes assumes Gaussian distributions per feature, which doesn't match pixel value distributions well.
- Not suitable for high-dimensional correlated data: With 784 correlated pixel features, the independence assumption is strongly violated.
- Limited feature interactions: Cannot capture complex relationships between pixels.

Impact of PCA on Classification Methods

PCA Configuration

PCA Configuration used:

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Components used: 131
Explained variance: 95.02%
Dimension reduction: 784 → 131 (83.3% reduction)
```

Fig 2. PCA Configuration Used.

Empirical Results

Results are as follows:

COMPREHENSIVE COMPARISON TABLE						
Classifier	Accuracy (No PCA)	Accuracy (With PCA)	Δ Accuracy	Train Time (No PCA)	Train Time (With PCA)	Training Speedup
Logistic Regression	0.9350	0.9375	0.0025	9.505056	4.930840	1.927675
K Nearest Neighbors	0.9450	0.9525	0.0075	0.005881	0.002657	397.204773
Support Vector Machines	0.9675	0.9700	0.0025	0.271566	0.098861	2.746961
Decision Trees	0.7500	0.7425	-0.0075	0.186109	0.234973	0.792044
Random Forest	0.9425	0.9175	-0.0250	0.227169	0.328224	0.692114
Naive Bayes	0.7200	0.8700	0.1500	0.043561	0.006224	6.999310

Fig 3. Results with PCA

Detailed Impact Analysis

Impact analysis w.r.t Accuracy and Speed is as given below:

1 ACCURACY IMPACT:

- ➡ Logistic Regression : $\approx$ MINIMAL CHANGE (+0.0025)
- ➡ K Nearest Neighbors : $\approx$ MINIMAL CHANGE (+0.0075)
- ➡ Support Vector Machines: $\approx$ MINIMAL CHANGE (+0.0025)
- ➡ Decision Trees : $\approx$ MINIMAL CHANGE (-0.0075)
- ⚠ Random Forest : $\downarrow$ DECREASED by 0.0250
- ✅ Naive Bayes : $\uparrow$ IMPROVED by 0.1500

2 SPEED IMPACT (Training):

- ⚡ Logistic Regression : 1.93x faster (Moderate improvement)
- 🚀 K Nearest Neighbors : 397.20x FASTER (Significant improvement)
- 🚀 Support Vector Machines: 2.75x FASTER (Significant improvement)
- 🐢 Decision Trees : 0.79x (Slightly slower)
- 🐢 Random Forest : 0.69x (Slightly slower)
- 🚀 Naive Bayes : 7.00x FASTER (Significant improvement)

Fig 4. Impact Analysis

Overall Summary

Key findings are as follow:

- Without PCA the best performer was SVM with accuracy of 96.75% and Training time of 0.2716s.
- With PCA also the best performer is SVM with accuracy of 97% and less training time too.
- Effect of PCA w.r.t speed was the best on KNN with speed improvement of about 397%.
- Best overall accuracy achieved with classification methods is 97% and method is SVM.